

# RUTWIK PATEL

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## EDUCATION

University of Southern California, Los Angeles, CA

August 2023 - May 2025

Masters of Science in Computer Science

GPA: 3.81/4.0

University of Mumbai, Mumbai, India

August 2019 - May 2023

Bachelor of Technology in Information Technology

GPA: 3.8/4.0

## EXPERIENCE

Software Engineer Intern

Sept 2025 - Dec 2025

*Sigma Computing*

New York, NY

- Orchestrated better decision-making with formula-based data visualizations, improving data clarity for 60+ orgs.
- Ensured four smooth feature launches by driving work from development through prod and resolving issues pre-release.
- Increased production stability by expanding test coverage, catching regressions early and reducing post-release issues.
- Collaborated with product and design to refine feature behavior and edge cases, reducing overall support tickets.

Software Engineer Intern

Jul 2024 - Aug 2025

*World Salon*

Los Angeles, CA

- Solved slow speaker sourcing by automating data extraction and outreach workflows, processing 39,000+ profiles e2e.
- Improved team efficiency through AI-driven speaker profiling and personalized outreach, enabling scalable sourcing.
- Addressed performance and scalability bottlenecks by modularizing core, payment services improving latency by 20%.
- Spearheaded Event creation workflow using MERN stack with JWT authentication, role-based access control.

Research Assistant

Feb 2024 - May 2025

*USC Marshall School of Business*

Los Angeles, CA

- Implemented automated web scraper using Selenium to collect 10 years of SP1500 sustainability reports (15K+ PDFs).
- Engineered RAG system with 10K+ vector embeddings and semantic search across 500GB+ energy transition data.
- Executed multi-threaded PDF pipeline with OCR attaining 30% faster extraction vs. prior single-threaded workflow.
- Created PostgreSQL database indexing 2M+ data points with React UI for researchers accessing processed reports.
- Automated Orbis database scraper for 1.8M+ companies using dynamic threading, reducing manual workload by 90%

Software Development Engineer Intern

May 2022 - Jul 2022

*Smart Consultant*

Mumbai, India

- Designed inventory management system for small retail businesses with Django REST API and PostgreSQL database.
- Delivered dashboard analytics showing sales trends, profit margins, and automated ABC analysis for 5000+ SKUs.
- Deployed on AWS EC2 achieving 200ms response times with S3/CloudFront CDN, supporting 15+ concurrent clients.

## PROJECTS (ENGINEERING)

Stock Insight Application ([Angular Website](#), [iOS App](#)) | **Angular, Swift, Express, MongoDB, GCP**

- Developed a full-stack stock trading platform with Finnhub and Polygon APIs to deliver accurate market data.
- Achieved 96% performance score through lazy loading, responsive Angular Material design, and optimized API caching.
- Built a complementary iOS app in Swift; replicated core web features for seamless cross-platform experience.

Cataract Detection with Explainable AI (XAI) ([Github](#)) ([Paper](#)) | **React, Python, CNN, XAI libraries**

- Led team of 3 to develop CNN-based cataract detection system achieving 97% accuracy with explainable AI integration.
- Integrated GradCAM for visualizing model decisions and localizing affected regions in eye images through heatmap.
- Designed a web interface for clinicians to upload eye scans and receive interpretable predictions with confidence scores.

RAG-based Q&A App ([Github](#)) | **Streamlit, LangChain, Chroma, OpenAI Embeddings**

- Created a RAG Q&A app with vector search and real-time retrieval, enabling sub-second search across private docs.
- Optimized the pipeline with parallel chunking and batching, reducing API calls by 30% and cutting 20% latency.

Fake News Detection | **Python, CNN-LSTM, BERT, RoBERTa**

- Developed and benchmarked multiple deep learning architectures for fake-news detection, achieving 93% accuracy.
- Achieved 94% precision, recall, and F1-score on large-scale WELFake and Kaggle datasets with 115K+ articles.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, Go, Swift | **Databases:** PostgreSQL, MongoDB, MySQL, Chroma

**Frontend:** React, Angular, Next, React Hooks, Redux, Zustand, HTML/CSS, Jest, Cypress, Tailwind, Material UI

**Backend & APIs:** Node.js, Express.js, Django REST, Flask, REST APIs, Microservices, FastAPI, Redis

**Cloud & DevOps:** AWS, GCP, Docker, CI/CD, Terraform | **ML/Data:** PyTorch, TensorFlow, LangChain

## PUBLICATIONS

- Published 2 IEEE papers: federated learning ([SICTIM 2023](#)) and cataract detection ([PuneCon 2023](#)).